



第1章：设置Python环境

以使用 uv 为例

安装 Python

```
wanghaoping — zsh — 80x24
Last login: Mon Feb  2 18:38:14 on ttys005
wanghaoping@wanghaopingdeMac-mini ~ % python3 --version
Python 3.13.5
wanghaoping@wanghaopingdeMac-mini ~ %
```

```
wanghaoping — zsh — 80x24
Removing: /Users/wanghaoping/Library/Logs/Homebrew/openssl@3... (64B)
Removing: /Users/wanghaoping/Library/Logs/Homebrew/ca-certificates... (64B)
Removing: /Users/wanghaoping/Library/Logs/Homebrew/python@3.11... (2 files, 2.6K
B)
==> Caveats
==> python@3.13
Python is installed as
  /opt/homebrew/bin/python3.13

Unversioned and major-versioned symlinks `python`, `python3`, `python-config`, `
python3-config`, `pip`, `pip3`, etc. pointing to
`python3.13`, `python3.13-config`, `pip3.13` etc., respectively, are installed i
nto
  /opt/homebrew/opt/python@3.13/libexec/bin

If you do not need a specific version of Python, and always want Homebrew's `pyt
hon3` in your PATH:
  brew install python3

`idle3.13` requires tkinter, which is available separately:
  brew install python-tk@3.13

See: https://docs.brew.sh/Homebrew-and-Python
wanghaoping@wanghaopingdeMac-mini ~ %
```

打开终端，使用命令 `python --version` 或 `python3 --version` 检查版本。

如果未安装Python或版本较旧，使用命令进行安装：

- Linux:

`sudo apt update`

`sudo apt install python3.13 python3.13-venv python3.13-dev`

- Mac:

`brew install python@3.13`

- Windows:

请从官方网站下载并运行安装程序：

<https://www.python.org/downloads/>

创建虚拟环境

```
wanghaoping — zsh — 80x24

See: https://docs.brew.sh/Homebrew-and-Python
wanghaoping@wanghaopingdeMac-mini ~ % pip install uv
zsh: command not found: pip
wanghaoping@wanghaopingdeMac-mini ~ % pip3 install uv
Collecting uv
  Downloading uv-0.9.28-py3-none-macosx_11_0_arm64.whl.metadata (11 kB)
  Downloading uv-0.9.28-py3-none-macosx_11_0_arm64.whl (19.8 MB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 19.8/19.8 MB 2.1 MB/s eta 0:00:00
Installing collected packages: uv
Successfully installed uv-0.9.28

[notice] A new release of pip is available: 25.1.1 -> 26.0
[notice] To update, run: pip3 install --upgrade pip
wanghaoping@wanghaopingdeMac-mini ~ % uv venv --python=python3.13
Using CPython 3.13.11 interpreter at: /opt/homebrew/opt/python@3.13/bin/python3.13
Creating virtual environment at: .venv
? A virtual environment already exists at `.venv`. Do you want to replace it? [y
✓ A virtual environment already exists at `.venv`. Do you want to replace it?
yes
Activate with: source .venv/bin/activate
wanghaoping@wanghaopingdeMac-mini ~ % source .venv/bin/activate
(wanghaoping) wanghaoping@wanghaopingdeMac-mini ~ %
```

- 安装uv

pip install uv. 或 pip3 install uv

- 创建虚拟环境

uv venv --python=python3.13

- 激活虚拟环境

source .venv/bin/activate

如果使用的是 Windows系统, 可能需要将上面的命令替换为 `source .venv/Scripts/activate` 或 `.venv/Scripts/activate`

可以执行 deactivate 命令来停用环境

安装包

```
wanghaoping — zsh — 80x24
(wanghaoping) wanghaoping@wanghaopingdeMac-mini ~ % uv pip install /Users/wanghaoping/Desktop/大模型网课/第1章：设置Python环境/requirements.txt
? `~/Users/wanghaoping/Desktop/大模型网课/第1章：设置Python环境/requirements.txt`
looks like a local requirements file but was passed as a package name. Did you
mean `-r /Users/wanghaoping/Desktop/大模型网课/第1章：设置Python环境/requirement
✓ `~/Users/wanghaoping/Desktop/大模型网课/第1章：设置Python环境/requirements.txt`
looks like a local requirements file but was passed as a package name. Did you
mean `-r /Users/wanghaoping/Desktop/大模型网课/第1章：设置Python环境/requirement
s.txt`? · yes
" Resolving dependencies...
Resolved 135 packages in 3.06s
Prepared 55 packages in 3m 12s
Installed 135 packages in 765ms
+ absl-py==2.4.0
+ anyio==4.12.1
+ appnope==0.1.4
+ argon2-cffi==25.1.0
+ argon2-cffi-bindings==25.1.0
+ arrow==1.4.0
+ asttokens==3.0.1
+ astunparse==1.6.3
+ async-lru==2.1.0
+ attrs==25.4.0
+ babel==2.18.0
```

在激活虚拟环境后，可以使用 uv pip 安装所需的包。

例如：

uv pip install numpy pandas 或

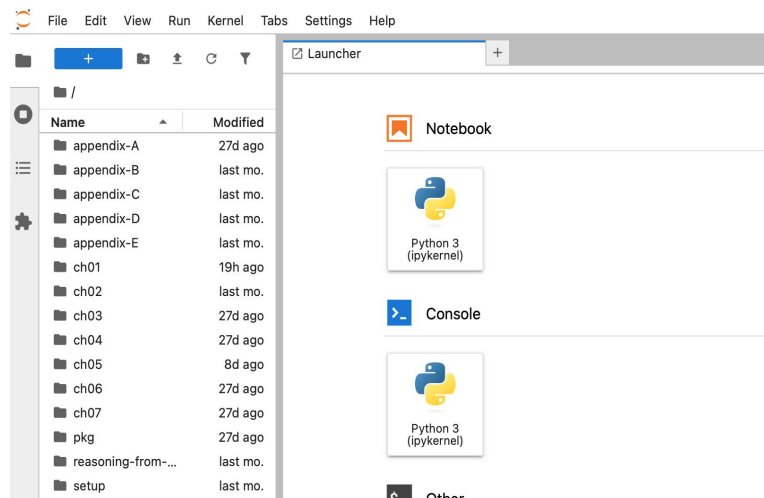
uv pip install **#your file path**

注：仓库提供了requirement.txt，可以直接下载好后续代码需要使用的包

开始写代码

```
LLMs-from-scratch-main — jupyter-lab — 80x24
d7e065cff5eb5482c09982c76981d8e4ba13af
[I 2026-02-03 13:14:49.766 ServerApp] http://127.0.0.1:8888/lab?token=6f43e6ef3dd7e065cff5eb5482c09982c76981d8e4ba13af
[I 2026-02-03 13:14:49.766 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2026-02-03 13:14:49.768 ServerApp]

To access the server, open this file in a browser:
file:///Users/wanghaoping/Library/Jupyter/runtime/jpserver-16181-open.html
Or copy and paste one of these URLs:
http://localhost:8888/lab?token=6f43e6ef3dd7e065cff5eb5482c09982c76981d8e4ba13af
http://127.0.0.1:8888/lab?token=6f43e6ef3dd7e065cff5eb5482c09982c76981d8e4ba13af
[I 2026-02-03 13:14:49.893 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server-nodejs, javascript-typescript-langserver, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, python-lsp-server, r-languageserver, sql-language-server, texlab, typescript-language-server, unified-language-server, vscode-css-languag eserver-bin, vscode-html-languageserver-bin, vscode-json-languageserver-bin, yam l-language-server
[I 2026-02-03 13:14:51.770 LabApp] Build is up to date
```



运行 `jupyter lab` 命令可直接启动 Jupyter Lab

常用的编译工具：

VScode: <https://code.visualstudio.com>

PyCharm: <https://www.jetbrains.com/pycharm/>

Jupyter Notebook: <https://jupyter.org>

Google colab: <https://colab.google>

Kaggle: <https://www.kaggle.com/code>

GitHub Codespaces: <https://github.com/features/codespaces>